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23R-D25 (PHY5204C)

2025

PHYSICS

(Major-II)

Paper : PHY5204C

(Classical and Relativistic Mechanics)

Full Marks : 45

Time : 2 hours

The figures in the margin indicate full marks for the questions.

1. Give very short answer : $1 \times 4 = 4$
- (a) What do you mean by generalised momentum ?
 - (b) What method do you adopt to change the domain from $(q, \dot{q}, t)$ to (q, p, t) in classical mechanics ?
 - (c) What is the physical significance of a differential equation of orbit in central force field ?
 - (d) Give an example of a cyclic coordinate for a Lagrangian of a physical system.

2. Give answer in brief :

2×3=6

(a) Express virtual displacement in generalised coordinate system.

(b) Write down two main features of motions in a central force field.

(c) Write down the Hamilton's action principle.

3. Answer the following questions : [(a) is **compulsory**, answer **either** (b) **or** (c), and **either** (d) **or** (e)]

5×3=15

(a) Show that d'Alembert principle is a different formulation of energy conservation principle.

(b) Derive Kepler's 2nd law for planetary motion.

OR

(c) The motion of the particle under the influence of a central force is described by $r = a \sin \theta$. Find the central force and corresponding potential function.

(d) Explain the concept of simultaneity in relativity.

OR

- (e) Derive the expression for mass energy equivalence in Special Relativity.

4. Answer **either** (a) **or** (b) : $10 \times 1 = 10$

- (a) Derive Lorentz space-time transformation equations for two inertial frames.

A particle with a mean proper life of 1 microsecond moves through the laboratory at a speed of 2.7×10^8 m/s, what will be its life time as measured by an observer in the laboratory?

$$7 + 3 = 10$$

- (b) Describe the Michelson-Morley Experiment. Point out the logical conclusions of the experiment.

5. Answer **either** (a) **or** (b) : $10 \times 1 = 10$

- (a) (i) Establish the relation $H = T + V$.
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- (ii) The particle is in motion in a central potential $V(r)$. Write down its Hamiltonian and hence establish its equation of motion.

$$2 + 4 = 6$$

(b) (i) Establish Hamilton's Canonical equations. 5

(ii) If the kinetic and potential energies are given by

$$T = \frac{1}{2} m \dot{r}^2 \text{ and } V = \frac{1}{r} \left(1 + \frac{r^2}{c^2} \right),$$

show that Hamiltonian

$$H = T + V.$$

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